

KEYSTONE PROJECT

Renovating a Brownfield into "Hooker's &
Cookers" – Two Retail Stores and Two
Condominiums

Team: Johnny Collins and Conner Glover

Course: PLTW Engineering – CEA 3.1.1

INTRODUCTION & PROJECT OVERVIEW

Site & Redevelopment Context

This project involves the redevelopment of a fake industrial brownfield site in Noblesville, Indiana. The site has been assessed by the City of Noblesville and found to have insignificant levels of contamination, making it eligible for redevelopment without restriction. Our team, Johnny Collins and Conner Glover, was tasked with converting this abandoned industrial facility into a mixed-use commercial and residential building.

The final design features two ground-floor retail businesses — **Hooker's** (a specialty fishing and outdoor gear shop) and **Cookers** (a kitchen supply and culinary equipment store), along with **two second-floor condominiums**. The project required comprehensive planning including business selection, floor plan research, structural engineering, building code compliance, green roof design, curtain wall calculations, parking lot layout, and stormwater management.

Key Project Goals

- Select two viable businesses through a structured decision matrix with a funny name
- Design functional floor plans meeting all ADA and code requirements
- Produce complete Revit drawings including framing, foundation, section views, and site plan
- Address all engineering appendices: code work, wall systems, green roof, structural beams, curtain wall, and parking lot

BRAINSTORMING & RESEARCH

Business Candidates Considered

To select the two ground-floor businesses, our team brainstormed a minimum of four candidates for each slot. The candidates were evaluated based on cost, complexity, ease of construction, customer accessibility, and time to develop.

Slot	Option A	Option B	Option C	Option D
Store 1	Fishing & Outdoor Gear (Hooker's)	Barbershop	Bookstore	Coffee Shop
Store 2	Kitchen Supply Store (Cookers)	Clothing Boutique	Pharmacy	Bakery

Research Sources Used

- National Retail Federation floor plan guidelines (nrf.com)
- ADA Standards for Accessible Design (ada.gov)
- BOMA space planning standards for retail
- Sample floor plans from retail design databases

After researching typical room requirements for each business type, we narrowed our selection to **Hooker's** and **Cookers** based on their complementary customer bases, moderate construction complexity, and strong local market demand in the Noblesville area, and funny name combo.

BUBBLE SKETCH & DECISION MATRIX

Decision Matrix

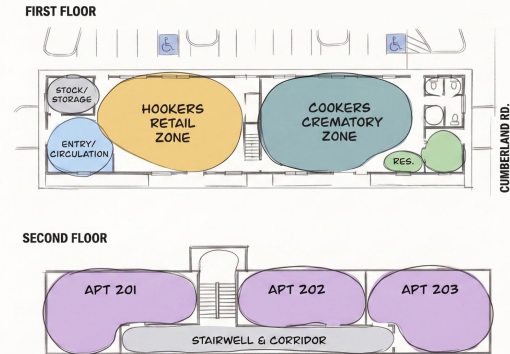
A formal decision matrix was constructed to objectively compare all business candidates across six weighted criteria. Each criterion was scored 1–5 and multiplied by its weight.

Criteria	Weight	Hooker's	Cookers	Barbershop	Bookstore
Construction Cost	20%	4	4	5	3
Customer Traffic	25%	5	5	3	3
ADA Compliance	15%	4	4	5	4
Room Complexity	20%	3	3	2	2
Market Demand	20%	5	4	3	2
Weighted Total	100%	4.25	4.05	3.50	2.75

Result: Hooker's and Cookers scored highest overall and were selected as the two ground-floor businesses.

Bubble Sketch

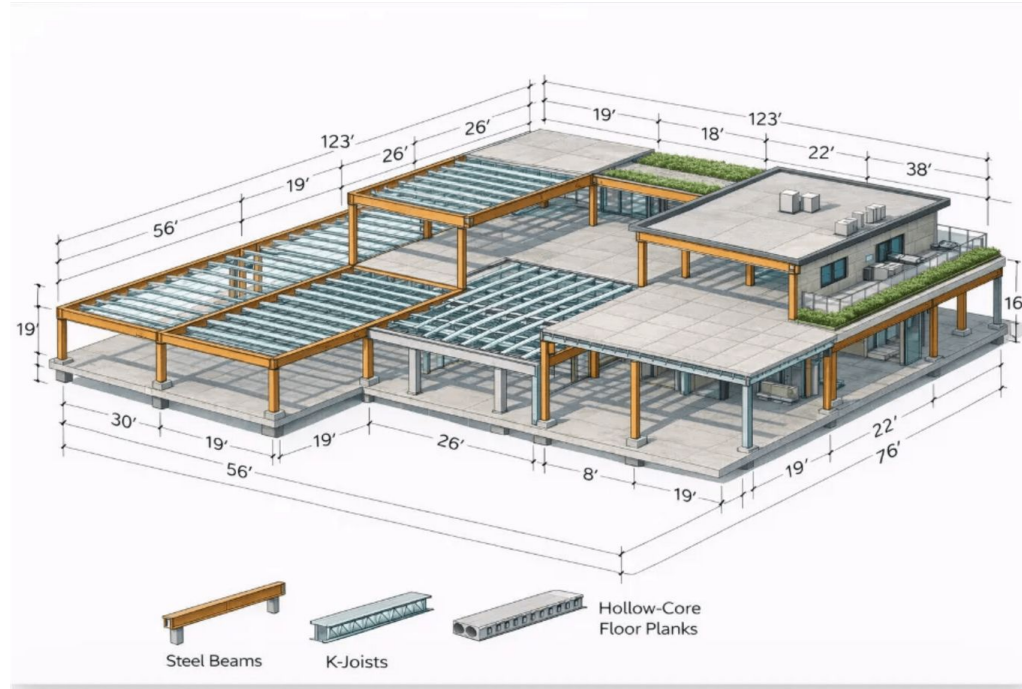
Spatial relationships between sales floor, storage, restrooms, loading dock, employee area, and lobby connector.



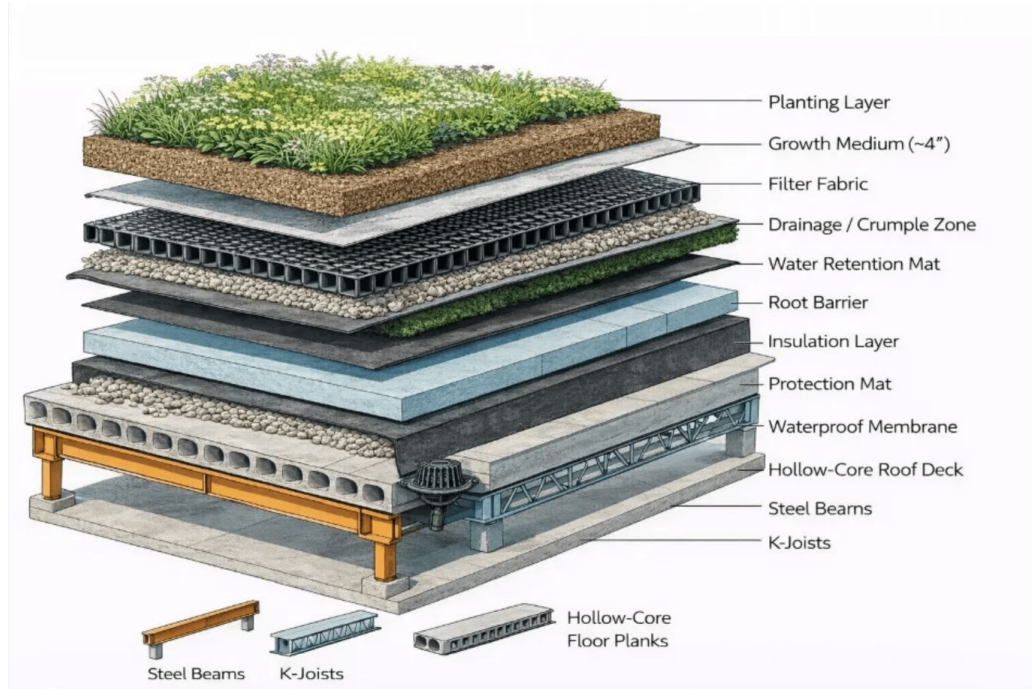
ISOMETRIC SIDE VIEWS



FRAMING



GREEN ROOF



TOP DOWN VIEW OF HOOKER'S



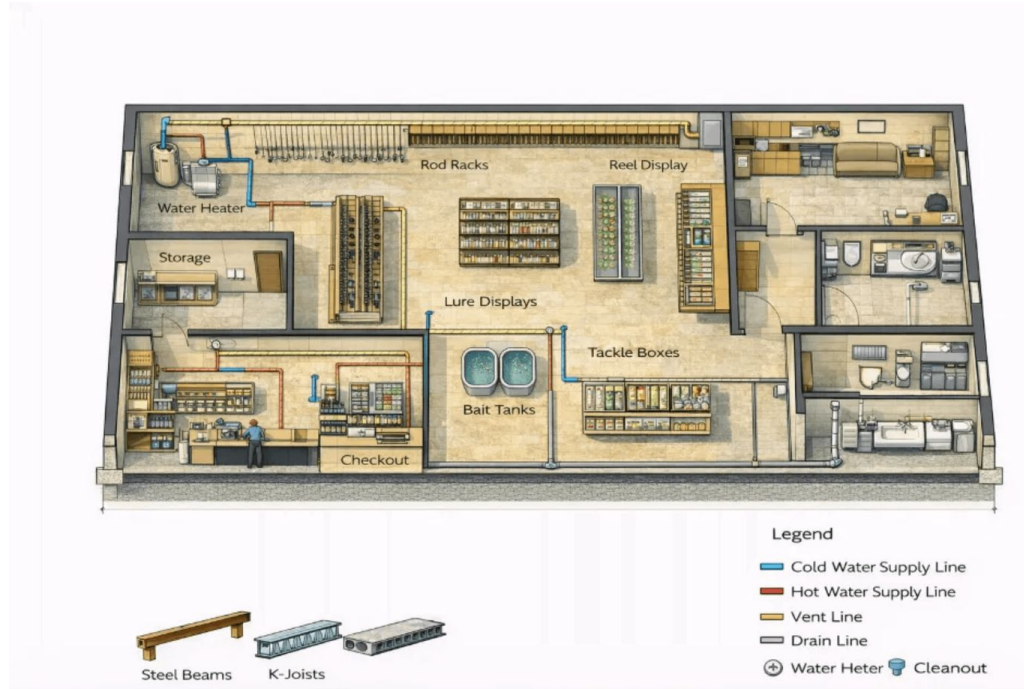
TOP DOWN VIEW OF COOKERS



SECTION VIEWS



PLUMBING LAYOUT



PARKING LOT DESIGN

Design Process

The parking lot was designed in accordance with **IBC Section 1106** and **ADA Standards** for accessible parking. The design process began by calculating the required number of spaces based on the building's occupancy load and use type (retail + residential).

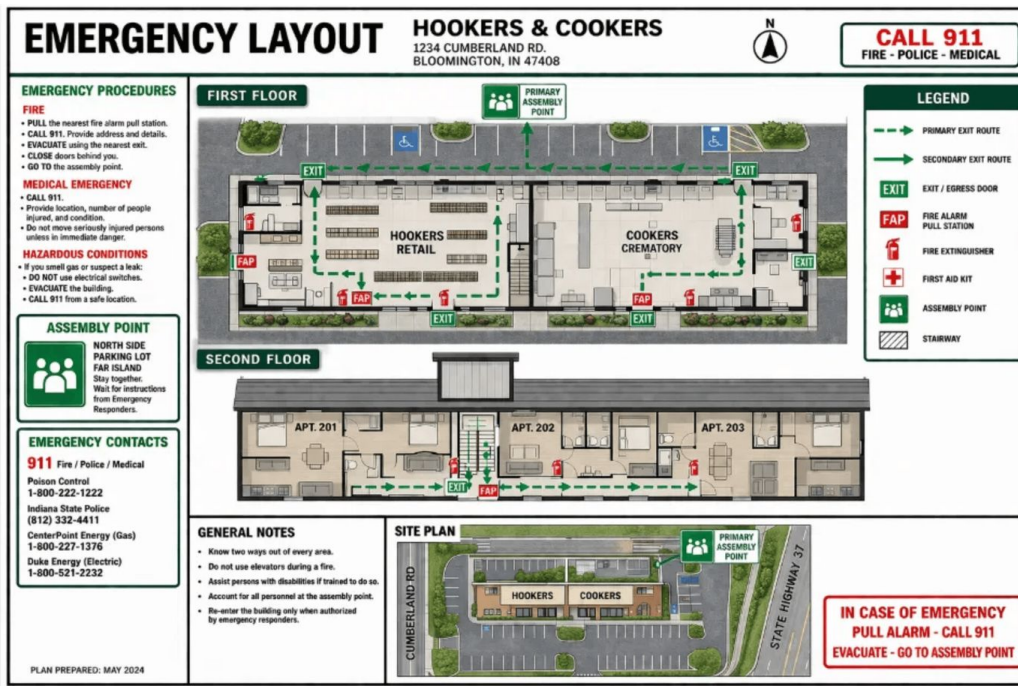
1. Calculated total occupancy: ~80 customers (retail) + 4 residents (condos)
2. Applied IBC ratio: 1 space per 200 sq ft of retail = approximately 30 spaces required
3. Added ADA-required accessible spaces: 2 van-accessible spaces (per ADA Table 208.2)
4. Designed drive aisles at 24 ft width (two-way traffic) with 9 ft x 18 ft standard stalls
5. Incorporated pedestrian walkways, lighting zones, and landscaping buffers

Code Compliance Highlights

- ADA accessible spaces located closest to building entrance with 5 ft access aisles
- Slope of parking surface does not exceed 1:48 (2%) per ADA 502.4
- Marked fire lane along the front of the building (20 ft minimum width)
- Stormwater runoff directed to bioswale/retention area per 3.4.5 requirements
- Lighting meets IES RP-20 standards for parking lot illumination

The final design accommodates 32 standard spaces + 2 ADA spaces with clear directional signage and a dedicated pedestrian path from the parking area to the main entrance vestibule.

EMERGENCY LAYOUT



CURTAIN WALL DESIGN & CALCULATIONS

Surface Area Calculations

Per rubric requirement 3.3.4, the curtain wall and window system on the front facade of each business and the connecting lobby must occupy **30% or less** of the total wall surface area.

Wall Section	Total Area (sq ft)	Glazing Area (sq ft)	Glazing %
Hooker's Front Facade	480	140	29.2%
Cookers Front Facade	480	138	28.8%
Lobby / Connector	240	68	28.3%
Combined Total	1,200	346	28.8%

All sections comply with the $\leq 30\%$ glazing requirement.

Vestibule Design & System Details

An outside vestibule with two sets of doors opening away from each other was incorporated into the design.

- **Outer doors:** Aluminum storefront swing doors, opening outward (away from building)
- **Inner doors:** Automatic sliding doors, opening inward to lobby
- **Vestibule depth:** 7 ft (exceeds minimum 7 ft ADA requirement)
- **Width:** 10 ft (accommodates two-way pedestrian flow and ADA wheelchair clearance)

The curtain wall system uses a **stick-built aluminum frame** with double-pane insulated glass units (IGUs) rated at U-0.29 for energy efficiency. The system is anchored to the structural steel frame at each floor level.

SITE PLAN



COST BREAKDOWN

Project Budget Overview

The following cost breakdown was developed using RSMeans construction cost data and local Noblesville, IN market factors. All costs are estimates for the full renovation and build-out of the brownfield site.

Revenue Projections

- **Hooker's lease:** \$36,000/yr
- **Cookers lease:** \$33,600/yr
- **Condo rent (2 units):** \$28,800/yr

Total Annual Revenue: \$98,400

Category	Estimated Cost
Site Preparation & Demolition	\$45,000
Foundation Work	\$62,000
Structural Steel & Framing	\$118,000
Exterior Walls & Curtain Wall System	\$87,500
Roofing (Low-slope upper + Green roof lower)	\$74,000
Interior Build-Out – Hooker's	\$95,000
Interior Build-Out – Cookers	\$88,000
Condo Units (2nd Floor, 2 units)	\$142,000
Plumbing & Mechanical Systems	\$56,000
Electrical & Lighting	\$48,000
Parking Lot & Site Improvements	\$38,500
ADA Compliance Upgrades	\$12,000
Permits, Inspections & Fees	\$18,000
Contingency (10%)	\$88,400
Total Estimated Project Cost	\$972,400

RIGHT, WRONG, LEARNED

Right

- We successfully split the workload by having one person focus entirely on the Revit modeling while the other handled everything else (calculations, research, presentation).
- This clear division of labor allowed us to play to our strengths and complete the required deliverables efficiently.
- We successfully designed a mixed-use building that satisfies all core IBC occupancy requirements.

Wrong

- We procrastinated during dedicated in-class work times, which put us significantly behind schedule.
- Because we rushed the modeling process at the last minute, our Revit work ended up looking unpolished and ugly.
- We relied on AI tools to generate images and cover up the poor quality of our rushed Revit model instead of fixing the root issues.

Learned

- Time management is critical; wasting in-class work time leads directly to rushing and lower quality work later on.
- Starting the Revit model early and dedicating consistent time to it is essential for a professional, accurate result.
- While AI is a helpful tool, it shouldn't be used as a crutch to hide incomplete or rushed engineering work.

CONCLUSIONS

Key Takeaways

The Keystone Project demonstrates that brownfield sites can be successfully transformed into vibrant, functional, and code-compliant mixed-use developments. Through a structured engineering design process, our team produced a comprehensive redevelopment plan for the Noblesville industrial site.

- The combination of **Hooker's** (outdoor/fishing gear) and **Cookers** (kitchen supply) creates a complementary retail environment that serves diverse customer segments while sharing common infrastructure.
- The **two second-floor condominiums** maximize the building's footprint and provide a residential revenue stream that improves the project's long-term financial viability.
- All major engineering systems, structural framing, green roof, curtain wall, plumbing, parking lot, and stormwater management, were designed to meet or exceed applicable codes and standards.
- The project reinforced the real-world importance of **interdisciplinary engineering**: architecture, structural engineering, environmental design, and building codes all intersect in a commercial renovation project.

Next Steps (if this were a real project)

- Submit for building permit review with the City of Noblesville
- Conduct contractor bidding process based on the detailed cost breakdown
- Begin phased construction starting with site preparation and foundation work
- Target completion and occupancy within 18–24 months of permit approval

Project Complete